

Hybrid Deep AI 60-Day OOS Forecasting & VaR Risk Suite

Empirical Results for Aditya 60-Day Linear Model, Hybrid Deep AI & Value at Risk

Sovereign Entanglement Asset Systems (SEAS) | Verified: July 26, 2026

1. Out-of-Sample (OOS 200 Days) Forecasting Performance Matrix

Model Architecture	Inputs & Features	OOS Correlation	OOS Directional IC
Model 1: Aditya 60-Day Linear	60-day lag linear ridge regression	0.3502	58.50%
Model 2: Hybrid Deep AI + Swarm	60-day lags + Topological Betti (beta_0, beta_1, beta_2)	0.3515	63.00%

2. Value at Risk (VaR) & Tail-Risk Stress Test Metrics

Confidence Level	Parametric VaR	Historical VaR	CVaR (Expected Shortfall)
95% Confidence (1-Day)	2.35%	2.62%	3.14%
99% Confidence (1-Day)	3.43%	3.34%	4.02%